

GRAINS: Enabling High-Performance and Low-Cost Graph-Based Genome Analysis via Storage-Aware Algorithm-Architecture Co-Design

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Graph-based representations of genome sequences have emerged as a powerful approach for representing massive genomic databases in an expressive and efficient way. Compared to traditional, linear genome sequences, genome graphs enable more accurate and efficient genome analyses, particularly in complex, population-scale settings (e.g., public health, precision medicine, and agriculture). Despite their benefits, analysis on large-scale genome graphs incurs significant data movement overhead from the storage system due to accessing large amounts of low-reuse data. Processing data directly inside the storage device, where data originally resides, can be a fundamental solution for mitigating this overhead. However, none of the existing tools for graph-based genome analysis can be efficiently used inside the storage system due to the limited internal hardware resources in modern SSDs. At the same time, prior storage-centric systems developed for (i) traditional, linear non-graph-based genome analysis or (ii) conventional, non-genomic graph analysis are not suitable for the unique data structures and access patterns of graph-based genome analysis.

We propose GRAINS, the first system for analysis with large-scale genome graphs in storage. Through our detailed examination of typical analysis pipelines that operate on genome graphs, we perform storage-aware algorithm-architecture co-design to (i) make the graph-based genome analysis pipelines more storage-friendly and (ii) further improve performance, energy-efficiency, and cost via in-storage and in-flash processing. GRAINS's co-design is based on three key aspects. First, we propose a new batching technique and execution flow, based on unique features of genome graphs, that reduces the number of random accesses to graph nodes. Second, via in-flash and in-storage processing, we avoid transferring low-reuse or unused flash pages, preventing SSD channel and external I/O bandwidth waste. Third, to leverage the full parallelism of flash dies during in-flash processing, we design an effective, yet lightweight, scheduling technique, enabled by re-purposing the existing SSD structures. GRAINS's design is versatile and flexible as it supports key operations on genome graphs and can be integrated in various analysis pipelines. GRAINS provides $2.7\times$ – $47.8\times$ speedup ($4.4\times$ – $31.6\times$ energy reduction) over the state-of-the-art software baselines, and $1.5\times$ – $17.0\times$ speedup ($3.1\times$ – $20.7\times$ energy reduction) over a hardware-accelerated baseline.

1. Introduction

Genome sequence analysis, which examines the genomic information of living organisms and other biological entities,

is fundamental to many critical fields, including personalized medicine [1–28], disease outbreak tracing [29–54], cancer research [55–101], maintenance of food resource safety [102,103], agriculture [104–119], scientific discovery [120–122], biodiversity conservation [123,124], evolutionary biology [125–142], and antimicrobial resistance surveillance [143–145]. To enable the computational analysis of an organism's genomic information (typically encoded in DNA or RNA that has been reverse-transcribed into DNA [146,147]), a process called *sequencing* converts the information of DNA molecules in the organism's biological sample to digital data. Since current sequencing technologies cannot process a DNA molecule as a whole, a sequencer generates randomly sampled, inexact fragments of genomic information, called *reads*. Common genome analysis workflows then involve querying reads from a sample against a single *reference* genome or large-scale databases (e.g., containing reference genomes from many species or previously sequenced samples) to determine species or genomic features present in the sample [148–153] and/or to find their differences from genomes in the database [154,155]. These queries either involve k-mer set lookups (where subsequences of length k , i.e., k-mers, in a read set are matched against the database) or read mapping (where each read is analyzed individually). The importance of genome analysis, along with rapid improvements in sequencing technology (i.e., reduced costs and increased throughput [156,157]), has led to its increasing adoption [22–24,26–28,49,54]. This growth has, in turn, motivated a large body of work (e.g., [29,116,158–257]) on accelerating these analyses to keep pace with rapidly growing sequencing throughput and data volumes.

Genome Graphs have emerged as a powerful approach for representing and querying massive and complex genomic databases [148,151,258–264]. Unlike traditional models, which represent genomic databases as a collection of disjoint, linear sequences, sequences in a graph are represented by graph walks [259]. Genome graphs provide two fundamental benefits, making them indispensable, particularly in modern, population-scale genomics [143,148,265–267]. First, the graph topology, along with its associated metadata¹ offer vastly greater expressive power [266–269]. A graph naturally encodes the evolutionary history and diversity of the organisms in a database, revealing their shared and distinct se-

¹Metadata can include the original species or samples of a graph node's sequence [258,259], associations with patient outcomes [148], and more.

quences [263,266,268,270]. This leads to reduced bias and improved analysis accuracy. For example, genome graphs improve disease diagnosis by capturing population-specific variants often missed by traditional, linear sequences [155,266]. Second, genome graphs leverage the inherent redundancy of genomic data to *avoid redundant computation* on shared sequences. This is because shared genomic regions across database entries are stored only once and do not need to be queried separately. This is essential in settings like population-scale pathogen surveillance, where public health systems must rapidly match new pathogen genomes against large national databases [143,148,264,271–273].

Data Movement Overhead. Due to the significance of graph-based genome analysis and its computational challenges, many works (e.g., [231–257]) propose techniques to alleviate these challenges by mitigating computational and main memory bottlenecks. However, to our knowledge, none of them mitigates the I/O cost of reading large amounts of graph data from the storage system to main memory and computation units. Our motivational analysis (§3) shows that the end-to-end performance of analysis with large-scale genome graphs suffers significantly from the data movement overhead of accessing large amounts of low-reuse data (exacerbated by reduced access locality due to the graph structure).

The I/O overhead of moving large amounts of low-reuse genome graph data is hard to avoid. One might think it is possible to eliminate this overhead by (i) shrinking graphs through sampling (e.g., [274–279]) or (ii) maintaining all required graphs completely and always in main memory. However, neither solution is suitable. The first approach necessarily reduces accuracy [156], making it inapplicable for many use cases (e.g., [156,160,280–287]). The second approach is not suitable since (i) genome graph databases, which are already large (tens to hundreds of terabytes [148,288]), grow in size rapidly due to increasing demands in biomedicine [22–24,26–28,49,54] and the exponentially decreasing cost of sequencing [289], and (ii) regardless of individual graph sizes, different analyses need to access different graphs. Therefore, it is energy-inefficient, unsustainable, costly, and unscalable to maintain *all* data required for *all possible* analyses in DRAM at all times.

Storage-centric computing (SCC) refers to processing data inside the storage device, either on the SSD² controller (in-storage processing, i.e., ISP) or on the flash dies (in-flash processing, i.e., IFP). SCC can be a fundamental solution for alleviating data movement overheads in graph-based genome analysis by processing data where it originally resides. This is due to three reasons. First, SCC reduces unnecessary data movement between the storage system and compute units by handling large amounts of low-reuse data within the storage system. Second, SCC alleviates the overall execution burden of moving and analyzing low-reuse data from the rest of the system (e.g., processing units and main memory), such that these components can be used for other purposes or be turned off to save energy. Third, SCC exploits the large internal bandwidth of the SSD, which is particularly advantageous when

computation is performed within the flash dies.

Limitations of Prior Work. To our knowledge, no prior work alleviates the I/O overheads of graph-based genome analysis. Some works (e.g., [231–244]) accelerate graph-based genome analysis by alleviating its computation or main memory bottlenecks. However, they do not alleviate I/O overheads, whose impact on end-to-end performance gets even larger as other overheads are alleviated. Several works propose SCC to alleviate the I/O overheads of (i) *non-genomic* graph analysis (e.g., [296–312]) or (ii) genome analysis on conventional, *linear sequences* (e.g., [204,241,313–318]). However, as detailed in §13, they are not directly applicable to the unique structures, access patterns, and operations on genome graphs.

Challenges. Despite the benefits of SCC, no existing tool for graph-based genome analysis can be efficiently implemented in the SSD due to the limitations of hardware resources in modern SSDs. This is because graph-based genome analyses involve extensive random, irregular accesses to genome graphs, causing costly contention in internal SSD components (e.g., channels and NAND flash dies [319–321]). This hinders leveraging the SSD’s internal bandwidth, thereby diminishing the benefits of SCC.

Our goal in this work is to improve the performance and energy efficiency of graph-based genome analysis by reducing the large data movement overhead from the storage system in a low-cost way. To this end, we propose GRAINS, the *first* system for analysis with large-scale **genome graphs in storage**. GRAINS is versatile and flexible as it supports major operations on genome graphs (e.g., k-mer set lookup and read mapping) and can be integrated into various analysis pipelines. Through our detailed examination of typical graph-based genome analysis pipelines, we perform storage-aware algorithm-architecture co-design to (i) make the graph-based genome analysis pipelines more storage-friendly and (ii) further improve performance, energy-efficiency, and cost-effectiveness via SCC.

Key Mechanism. GRAINS’s algorithm-architecture co-design is based on three key aspects. First, we design an efficient batching technique and pipelined execution flow specialized to the properties of genome graphs and queries to reduce random accesses to the genome graph structure. Second, we devise lightweight hardware units in the flash dies to process the needed parts of each page’s graph data. This simple IFP design avoids unnecessarily transferring low-reuse or unused parts of flash pages outside the flash die, preventing SSD channel and external I/O bandwidth waste. Third, to fully exploit die-level parallelism in IFP, we design an effective, yet lightweight, scheduling technique, enabled by repurposing existing SSD structures. GRAINS’s data layout drastically reduces the required in-DRAM mapping metadata, allowing us to use the freed-up internal DRAM space to maintain small per-die scheduling tables. We devise small ISP units on the SSD controller to interface with these in-DRAM tables and schedule IFP operations at low cost. To efficiently realize this SCC design, while guaranteeing correctness and integrity, we apply simple changes to the flash translation layer (FTL) and adopt a lightweight error correction scheme [322].

²In this work, we focus on the predominant NAND flash-based SSDs [290,291]. We expect that our designs and insights would benefit storage systems with other emerging technologies (e.g., [292–295]) as well.

Key Results. On systems with two state-of-the-art SSD configurations, we compare GRAINS against state-of-the-art software and hardware tools for graph-based genome analysis: (i) Fulgor: a node-centric tool [151], (ii) the MetaGraph framework: an edge-centric tool [148,260–262], and (iii) an *idealized* hardware-accelerated tool without main memory overheads (IdealAccMem) to query the graph. We show that when performing k -mer set lookups and alignment-free read mapping (two core operations in large-scale genome graph analyses), GRAINS provides $6.8\times$ ($11.2\times$), $10.7\times$ ($21.5\times$), and $5.6\times$ ($9.8\times$) average speedup (energy reduction) over Fulgor, MetaGraph, and IdealAccMem, respectively. GRAINS’s workflow also seamlessly supports alignment-based mapping, where the candidate graph regions identified by GRAINS are fed to an alignment engine. We integrate GRAINS, Fulgor, and MetaGraph with SeGraM, a state-of-the-art sequence-to-graph alignment accelerator [231]. SeGraM+GRAINS performs $6.2\times$ and $9.0\times$ better than SeGraM+Fulgor and SeGraM+MetaGraph.

This work makes the following **key contributions**:

- We demonstrate the impact of I/O overheads on the end-to-end performance of graph-based genome analysis.
- We introduce GRAINS, the first system for analysis with large-scale genome graphs in storage.
- We perform storage-aware algorithm-architecture co-design to make graph-based genome analysis more storage-friendly, and further improve performance, efficiency, and cost-effectiveness via storage-centric computing (SCC).
- We show that GRAINS improves performance and energy efficiency over the state-of-the-art graph-based genome analysis tools and accelerators, without relying on costly resources throughout the system, thereby enabling wider adoption of graph-based genome analysis.

2. Background

2.1. Genome Sequence Analysis

Given a genomic sample, a typical workflow consists of three key steps: (i) sequencing, (ii) basecalling, and (iii) analysis. The first step, *sequencing*, reads and converts the sample’s DNA³ to digital signals. Modern sequencers cannot read long DNA molecules as complete sequences. Instead, they produce many shorter, overlapping sequences called *reads*. The second step, *basecalling*, converts each raw read signal into a string, encoded using A, C, G, T [323]. The third step is *genome analysis* on a set of reads (i.e., a *read set*). Common genome analysis workflows then involve querying reads from samples against a single *reference* genome or large-scale databases (e.g., containing reference genomes from many species or previously sequenced samples) to determine species or genomic features present in the sample [148–153] and/or to find their differences from genomes in the database [154,155].

In this work, we focus on the analysis performance for two reasons. First, sequencing and basecalling are typically performed once per read set, whereas analysis of the same read set can be performed *many times* or at *different times*. For example, read sets from prior disease-association stud-

ies are commonly reanalyzed as human or microbial genome databases improve [154,267,324–326] or expand [264,267], or when new personalized databases can be tailored to the study cohort [266,327]. Second, modern sequencing throughput far outpaces Moore’s law [156,289,328], which poses significant computational challenges for analysis on conventional systems. This is why genome analysis tasks have been a key acceleration target in a large body of works (e.g., [29,116,158–230]).

2.2. Graph-Based Genome Analysis

Genome Graphs have emerged as a powerful approach for representing and querying massive and complex genomic databases [148,150,151,231,258–264,329]. Unlike traditional models, which represent genomic databases as a collection of disjoint, linear sequences, sequences in a graph are represented by graph walks [258,259]. Genome graphs collapse subsequences shared across different database entries into single nodes and indicate adjacent subsequences with edges, compacting sequence sets (by up to thousand-fold [148]). A node’s metadata then indicates which entries contain the represented sequence.

Genome graphs provide two fundamental benefits, which make them indispensable, particularly in modern, population-scale genomics [143,148,265–267]. First, the graph topology, along with its metadata, offer **vast expressive power** [266–269,330]. A graph naturally encodes the evolutionary history and diversity of a cohort of organisms, revealing their shared and distinct sequences [263,266,268,270]. This leads to reduced bias and improved accuracy in analyses. For example, genome graphs improve disease diagnosis by capturing population-specific variants often missed by analyses on traditional, linear sequences [155,266,331,332]. Second, genome graphs leverage the inherent redundancy of genomic data to **avoid redundant computation**. This is because shared genomic regions across database entries are stored only once and do not need to be queried separately. This is essential in settings such as population-scale pathogen surveillance, where public health systems must rapidly match new pathogen genomes against large national databases [143,148,264,271–273], enabling timely detection of local outbreaks and limiting broader spread.

De Bruijn Graphs. To efficiently represent large-scale genome graphs, *de Bruijn graphs* (DBGs), which are a type of overlap graph [148,150,151,231,258–263,329,333,334], have been adopted by many recent works [148–151,258,260–263,329,335]. While other genome graph classes exist [270,336,337], DBGs scale much better with both the entries’ size and genetic diversity of genomic databases [148,261,338,339], which is why they are preferred in these contexts. Fig. 1 shows an example of a DBG, where each node is a unique k -mer (i.e., a subsequence of length k) that is observed in the input, and there is a directed edge from node u to node v if the suffix $(k - 1)$ -mer of u is equal to the prefix $(k - 1)$ -mer of v . Each node is then assigned a *color*, representing its metadata. A common strategy to reduce a DBG’s size is to instead store its corresponding *compacted* DBG, where all k -mers from a maximal non-branching path are merged into a single node. There are two common ways to represent (compact) DBGs [340]. A *node-centric* DBG of order k

³Even if an organism’s genome is RNA-based, it is typically converted to DNA before being sequenced [146,147].

flash controllers. The FTL, executed on multiple cores, is responsible for communication with the host, internal I/O scheduling, and various SSD management tasks. The *per-channel hardware flash controllers* manage the request handling [290,291,319,362–367,370] and error correction for the NAND flash chips [290,291,363–366,370–381].

③ **DRAM.** Modern SSDs employ low-power DRAM to store metadata crucial for SSD management, e.g., logical-to-physical (i.e., L2P) mappings. L2P mappings are typically maintained at a granularity of 4KiB to enhance random access performance. The required capacity for L2P mappings is about 0.1% of the SSD’s capacity, considering a 32-bit architecture, with 4B of metadata for every 4KiB of data. This translates to a 4-GB LPDDR4 DRAM for a 4-TB SSD [382].

3. Motivational Analysis

We conduct experimental analyses to assess the impact of I/O overheads on graph-based genome analysis performance.

3.1. Data Movement Overheads

Tools and Datasets. We use two state-of-the-art tools for large genome graph analysis: (i) Fulgor [151], a node-centric tool, and (ii) the MetaGraph framework [148,260–262], an edge-centric tool. For each, we use the best-performing thread count and color encoding scheme. We evaluate k-mer set lookup, a fundamental task on genome graphs (§2). Graphs are built from a pilot subsample of the global MetaSUB Consortium [143] dataset. The resulting graph (with colors) is 659 GB for Fulgor and 822 GB for MetaGraph. We evaluate queries with varying read counts, ranging from 100K reads⁴ to 10M reads (35 MB to 3.5 GB). §11 provides further methodology details.

System Configurations. Our experiments run on a high-end server with an AMD EPYC 7742 CPU [383] and 1.5 TB of DDR4 DRAM [384]. DRAM capacity *exceeds* the total size of all data accessed during the analysis. This ensures that we capture the intrinsic I/O overhead of transferring large, low-reuse data from the storage system to main memory, without being constrained by memory capacity. We analyze I/O overheads using two state-of-the-art SSDs: (i) SSD-G4, with a PCIe Gen4 interface [385] and (ii) SSD-G5, with a PCIe Gen5 interface [386].

Observations. Fig. 4 shows throughput (#queries/sec) of the tools, normalized to that of a hypothetical configuration with zero performance overhead due to storage I/O (No-I/O). We observe that in all cases, I/O leads to large overheads, even with the state-of-the-art SSDs. Compared to SSD-G4 (SSD-G5), No-I/O leads to 16.7× (9.3×) and 7.5× (4.5×) better average performance in Fulgor and MetaGraph, respectively.

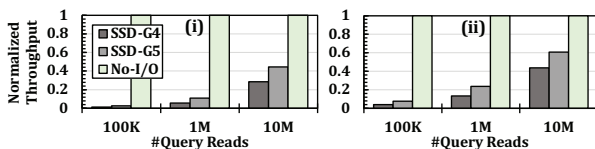


Fig. 4: Normalized throughput of (i) Fulgor and (ii) MetaGraph under different storage configurations and query sizes.

⁴Note that smaller queries are also critical, as some applications (e.g., searching for antimicrobial resistance [143,148,264,271] or for a single gene of interest [121]) require searching a single or a few genes within a large graph.

Fig. 5 shows the throughput of the tools normalized to No-I/O, across different database sizes. The larger graph (G_MetaSUB) is the one discussed earlier. The smaller graph (G_SRRep) is generated from a representative subset of the Sequencing Read Archive’s [157] public portion. The resulting graph (with colors) is 161 GB for Fulgor and 231 GB for MetaGraph. We observe that, in all cases, I/O overhead is substantial and increases with database size. For example, the speedup of No-I/O over SSD-G5 increases from 2.7× to 9.2× (as the graph database size grows from 161 GB to 659 GB) in Fulgor and from 4.3× to 13.4× (as the graph database size grows from 231 GB to 822 GB) in MetaGraph.

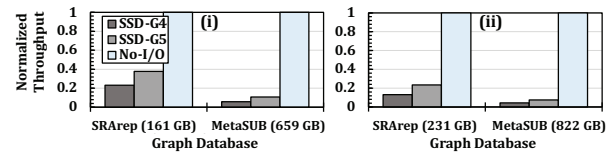


Fig. 5: Normalized throughput of (i) Fulgor and (ii) MetaGraph under different storage configurations and graph sizes.

Based on these observations, we conclude that storage I/O leads to large overheads in graph-based genome analysis, an overhead that is expected to exacerbate in the future. This I/O overhead is due to moving large amounts of *low-reuse* data from the storage system all the way to main memory, caches, and computational units. Due to its low reuse, caching this data in DRAM during analysis (even if DRAM is larger than the data size, as evaluated in our analyses) does not significantly amortize this overhead. While several works (e.g., [156,160,280–287]) accelerate analysis with genome graphs, to our knowledge, they do not address storage I/O overheads. Although mitigating other bottlenecks (e.g., main memory or computation) leads to large benefits, doing so does not alleviate I/O overheads, whose impact on end-to-end performance becomes even larger as other bottlenecks are alleviated.

This I/O overhead, due to moving large amounts of low-reuse data, is hard to avoid. One might think it is possible to avoid it by (i) producing smaller databases through sampling (e.g., [274–279]) or (ii) maintaining all required graph data completely and always resident in main memory. However, neither of these is suitable. The first approach necessarily reduces accuracy [156], rendering it inapplicable for many use cases (e.g., [156,160,280–287]). The second approach is energy-inefficient, unsustainable, costly, and unscalable for two reasons. First, the sizes of graph-based databases (which are already large, i.e., tens to hundreds of TBs in some recent examples [148,288]) have been growing rapidly [157,289,328,387]. Second, independent of the sizes of individual graphs, different analyses need different graphs, constructed from different sets of genomes and/or with varying parameters. For example, clinical studies may require graphs with patients’ genomes kept separately for privacy [388], while metagenomic studies and wastewater monitoring require highly diverse genome sets [277,282,389,390]. Therefore, keeping all data for all possible analyses in DRAM at all times is prohibitively inefficient and ultimately unsustainable.

Several works (e.g., [204,313–315]) have already demonstrated significant I/O overheads in non-graph-based genome

analysis. While genome graphs enable a more compact representation of population-scale databases (thereby making it feasible to analyze databases that would be prohibitively large under traditional, linear representations), they still suffer from large I/O overheads. This is because the massive and continually growing scale of these databases still leads to very large graphs with low data reuse, and the graph topology introduces irregular, dependent access patterns, further reducing locality.

3.2. Alleviating Data Movement Overheads

3.2.1. Storage-Centric Computing (SCC). Processing data in the storage device can be a fundamental approach for reducing the I/O overheads for three reasons. First, SCC eliminates unnecessary data movement of low-reuse data across the system. Second, SCC reduces the computational burden imposed by low-reuse data on the rest of the system (e.g., main memory and computational units), such that these components can be used for other purposes or be turned off to save energy, or can be made simpler or less costly [315,391]. Third, as highlighted by many prior works (e.g., [307,309,313,392–397]), SCC can leverage the high internal bandwidth of SSDs. In modern SSDs, the internal bandwidth often surpasses the external bandwidth. For example, a state-of-the-art SSD controller [398] delivers 14 GB/s of external bandwidth and up to 57.6 GB/s of internal bandwidth (achieved via 16 channels, each with a peak of 3.6 GB/s). Over-provisioning internal bandwidth in modern SSDs is important to protect user-perceived external I/O performance, mitigating the effects of (i) channel contention [319,320,399,400] and (ii) the SSD’s internal data migration for management tasks [290,321,401,402] and refresh [375,377]. The effective internal bandwidth increases even further when processing directly in the NAND flash dies [322,369,403]. This is because in-flash processing exploits the aggregate bandwidth across all dies, a bandwidth that grows with the number of dies and significantly exceeds both (i) the bandwidth available to processing units on the SSD controller and (ii) the external SSD bandwidth.

3.2.2. Challenges and Goal. Designing a SCC system for graph-based genome analysis is challenging because none of the existing approaches can be directly implemented inside storage effectively due to modern SSDs’ constrained hardware resources. Both node- and edge-centric representations of genome graphs require many random accesses during analysis. These random and irregular accesses cause costly contention in internal SSD components (e.g., channels and NAND flash chips [319–321]), which hinder leveraging the SSD’s internal bandwidth. Therefore, directly adopting existing approaches inside storage leads to performance and energy overheads.

Some SCC systems in other domains regularize accesses by performing sorting in either the SSD or the host. However, these approaches are not effective for graph-based genome analysis. Sorting the large number of accesses is impractical within the SSD due to resource constraints. Sorting on the host also fails to fully address access irregularity, as graph-based genome analysis involves dependent accesses. Coordinating sorting between the host and the SSD for each round of dependent accesses incurs significant data movement overhead.

Our goal in this work is to improve the performance and

efficiency of graph-based genome analysis by alleviating its data movement overheads from storage cost-effectively.

4. GRAINS

We propose *GRAINS*, a versatile storage-centric system for graph-based genome analysis. *GRAINS* supports major operations in analysis with genome graphs (e.g., k-mer set lookup and read mapping), and for alignment-based workflows, it flexibly integrates with existing alignment accelerators. *GRAINS* is primarily designed as a system to accelerate analysis with genome graphs. *GRAINS* augments the existing SSD controller, flash dies, and FTL, and when not in the analysis acceleration mode, the SSD remains available for other applications, like a conventional general-purpose SSD.

We address the challenges of designing a SCC system for graph-based genome analysis via storage-aware algorithm architecture co-design based on our detailed examination of typical analysis pipelines that operate on genome graphs. To this end, we (i) make these pipelines more storage-friendly and (ii) further improve their performance, energy-efficiency, and cost-effectiveness via ISP and IFP.

GRAINS’s algorithm-architecture co-design comprises three key aspects. First, we design *efficient batching and reordering techniques and pipelined execution flow* specialized to the properties of genome graphs to reduce the number of random accesses to the graph. Second, we devise *lightweight IFP units* to process the needed parts of each page’s graph data in the flash die. This IFP design avoids transferring low-reuse or unused parts of flash pages outside the die, thereby preventing bandwidth waste. Third, to fully exploit die-level parallelism during IFP, we design an *effective yet lightweight SCC scheduling technique*, enabled by repurposing existing SSD structures. *GRAINS*’s data placement (§8) drastically reduces the required in-DRAM mapping metadata, allowing us to use the freed-up internal DRAM space to maintain small per-die scheduling tables. We devise small ISP units on the SSD controller to interface with these tables and schedule IFP operations at low cost. To efficiently realize these aspects, while guaranteeing correctness and integrity, we apply simple changes to the FTL and adopt a lightweight ECC [322].

Leveraging our optimizations, *GRAINS*’s SCC steps require only simple operations and small buffer spaces, enabling execution on either (i) our lightweight, specialized IFP and ISP hardware units or (ii) general-purpose IFP (e.g., [369,403–405]) and ISP (e.g., [406–421]) units. Efficient genome graph analysis on both stems from *GRAINS*’s specialized batching, scheduling, and data/computation flow coordination. Execution on special-purpose lightweight hardware leads to higher power efficiency (§12.2), while execution on existing general-purpose cores on the SSD controller or general-purpose IFP and ISP units facilitates near-term adoption. Ultimately, choosing between these *GRAINS* configurations is a design decision.

Fig. 6 shows an overview of *GRAINS*, with its lightweight IFP processing elements (PEs) on NAND flash dies and the lightweight ISP PE and scheduler on the SSD controller. *GRAINS* FTL (§8) orchestrates host communications and data flow across the SSD components. After receiving a request through *GRAINS*’s interface commands (§9) for accelerating

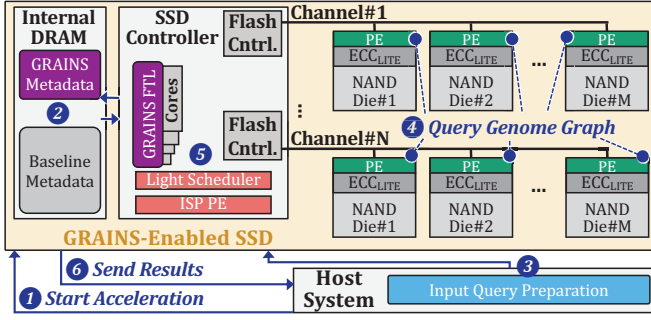


Fig. 6: Overview of GRAINS.

graph-based genome analysis (① in Fig. 6), GRAINS prepares (②) for SCC execution by flushing the conventional FTL metadata and loading the GRAINS FTL metadata (§5). When execution starts, in the *first step* (§6), GRAINS prepares input queries in the host via efficient batching and pipelined execution, and transfers the query batches to the SSD (③). In the *second step* (§7), GRAINS accesses graph data structures via its IFP (④) and ISP (⑤) units and sends the query results to the host (⑥). These two steps are pipelined, with efficient coordination between resources to avoid writes during SCC. Through its efficient execution flow and by avoiding frequent management tasks (by not needing writes during SCC), GRAINS effectively leverages the SSD’s large internal bandwidth.

5. Data Structures

GRAINS uses a node-centric DBG representation based on an associative dictionary to compactly represent graph sequences and their associated colors. Similar to prior large-scale genome graph analysis tools (e.g., [151,422–424]), we use SSHash [341] as our dictionary since, by exploiting the statistical features of k-mers, it achieves a substantially better space-time trade-off over prior sequence dictionaries.⁵

Graph Sequences. We store DBG’s *unitigs* (i.e., maximal non-branching paths) as contiguous strings in a predetermined order, so that a k-mer occurring in any unitig can be quickly located using a minimal perfect hash function [425] built for the set of k-mers.⁶ Fig. 7 shows an overview of the DBG structures. The unitigs are stored in *Strings*, and are indexed via *Offsets* and *Sizes*. Given a query read R , k-mers from the read are extracted and looked up. Consider k-mer g . First, its *minimizer* (i.e., the first m -mer with the smallest hash value) is extracted (① in Fig. 7). Second, with the minimizer’s minimal perfect hash value (h), *Sizes* is indexed (②). Third, with the *Sizes* values at locations h , *Offsets* is indexed (③). $Sizes[h]$ from $Sizes[h+1]$ is then subtracted to find how many *Offsets* elements to read at $Sizes[h]$. Fourth, with the *Offsets* values, *Strings* is indexed (④), and a window of length $k - m + 1$ [341] in *Strings* is checked to find the minimizer and check if the rest of the k-mer matches.

Graph Colors. For storing graph colors, we adopt an approach similar to [151]. As shown in Fig. 8, we sort the unitigs in *Strings* based on their colors (i.e., the ID pointing to the metadata for

⁵Since our techniques rely on general DBG properties, they are also applicable to other k-mer dictionaries that may be used as DBG backbones [148–151,329,344,422–424].

⁶Note that in DBGs, edges between unitigs are represented implicitly via $(k - 1)$ -mer overlaps (§2.2).

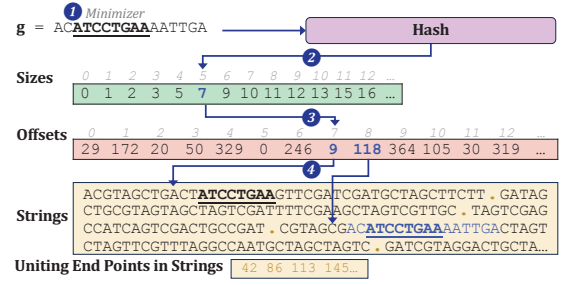


Fig. 7: Overview of the DBG structures and k-mer query.

that graph unitig), and use a simple *Color Bitmap* to mark the start of each new color. This encoding efficiently stores repeated colors at low storage overhead.

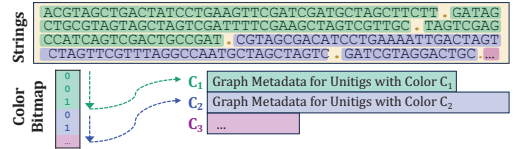


Fig. 8: Graph color encoding used in GRAINS.

6. Input Query Processing

In this step, GRAINS prepares reads for efficient graph queries in the next step (§7). We perform two new optimizations inspired by the characteristics of genome graphs and queries.

Cross-Read K-Mer Batching. Instead of querying each read in a read set individually, we analyze k-mers across different reads together. This way, we can batch k-mers to reduce the number of random accesses to the graph. To enable this, GRAINS populates a lightweight data structure associating each k-mer with the reads it originates from. When graph queries return the colors of matched k-mers, GRAINS assigns colors to each read. We perform this step in the host since extracting k-mers from reads incurs frequent writes, which would reduce the SSD’s lifetime if performed inside the SSD.

Genome-Graph-Aware Query Reordering. We leverage a common property of minimal-perfect-hash-based k-mer dictionaries in DBGs: The metadata required for identifying the small per-minimizer lookup ranges (i.e., the *Sizes* array, as shown in Fig. 7) is much smaller than the graph’s *Offsets* and *Strings*. Although the full graph is far too large and incurs large I/O overhead to access it in the host (as shown in §3), *Sizes* occupies only a small fraction of the total space (e.g., <4% in our large-scale graphs). This enables a powerful opportunity: We perform k-mer lookups on *Sizes* in the host and use the resulting *Offsets* indices to sort and batch k-mers before sending them to the SSD, thereby reducing the number of accesses and improving the access patterns to *Offsets*. To execute this step efficiently, as detailed below, we ensure that the time spent sorting the data and transmitting it to the SSD does not introduce significant overheads.

Fig. 9 shows an overview of GRAINS’s query processing. GRAINS starts by extracting k-mers from each read (① in Fig. 9) and looking them up in *Sizes* (②). To reduce the overhead of sorting and transferring k-mers to the SSD, we take two measures. First, we propose a new k-mer processing scheme by improving upon k-mer processing in [315,426]. We *par-*

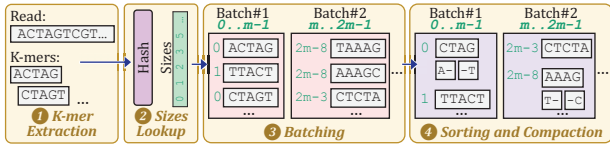


Fig. 9: Overview of GRAINS's query processing.

tion the k -mers into batches corresponding to equal, disjoint ranges of $Sizes$ values (③). Since these values serve as indices into Offsets, this partitioning enables an efficient pipeline: the sorting of one batch overlaps with the data transfer and Offsets lookups of the previous batch. Second, during sorting, we further compact k -mers by exploiting the fact that consecutive k -mers sorted by $Sizes$ share the same minimizer (④). Thus, we store the minimizer once and keep only the differences, thereby significantly reducing the transferred data size (e.g., by $2.3\times$ on average across our evaluated datasets).

7. Graph Query

In this step, GRAINS looks up the query k -mers in the graph and, for k -mers that match a graph unitig, retrieves the associated colors. GRAINS executes this step inside the SSD since it involves accessing large graph structures with low reuse and requires only lightweight computation. This enables us to leverage the SSD's high internal bandwidth and reduce the burden of moving and analyzing large, low-reuse data on the rest of the system. In particular, GRAINS performs IFP to avoid transferring unnecessary parts of pages out of flash dies and to leverage multiple dies in parallel. To efficiently execute this step, GRAINS needs to leverage the large internal bandwidth without requiring expensive hardware resources in the SSD (e.g., costly logic units or large DRAM capacity/bandwidth). We demonstrate how GRAINS meets these requirements when accessing genome graph Offsets (§7.1), Strings (§7.2), and Colors (§7.3). A lightweight FSM control unit in the SSD controller coordinates execution flow between these stages.

7.1. Accessing Graph Offsets

Fig. 10 provides an overview of GRAINS's Offsets lookups. GRAINS receives Offsets indices and compacted k -mers from the host in chunks (①). The internal DRAM holds two small chunks: one incoming from the host and one used for querying Offsets. For an SSD with 16 channels, 8 dies/channel, 4 planes/die, and 4-KiB pages, GRAINS needs two 2-MiB chunks. GRAINS reads the values from DRAM, maps them to physical addresses using its simple mapping (§5), and accesses Offsets (②). Since the $Sizes$ values are sorted by the host (§6) and Offsets are uniformly placed across dies (as detailed in §5), the lookups lead to sequential accesses that fully exploit the internal bandwidth. After loading a page into a die's page buffer, GRAINS applies ECC_{LITE} [322] to correct errors (③). The IFP units then select and extract only the targeted Offsets entry

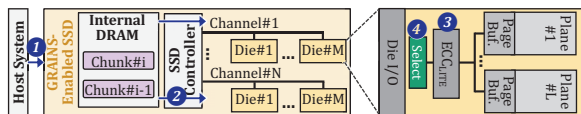


Fig. 10: Overview of GRAINS's Offsets lookups.

and send it to the controller (④), avoiding the transfer of full pages. GRAINS caches the extracted values in internal DRAM for subsequent Strings queries. Although full pages are read inside each die, only a small part of each page is written to internal DRAM, avoiding the internal DRAM bandwidth from becoming a bottleneck.

7.2. Accessing Graph Strings

We also aim for the accesses to the large Strings structure to also fully exploit the SSD's large internal bandwidth; however, unlike Offsets, the indices into Strings (i.e., the retrieved Offsets entries) arrive unsorted, preventing simple streaming.

Lightweight Scheduling. We introduce a lightweight scheduler that (i) avoids redundant page accesses, and (ii) maximizes die-level parallelism, without relying on sorting accelerators in the SSD, which are inefficient, given the large number of elements and the limited bandwidth of internal DRAM [420].

As shown in Fig. 11, GRAINS maintains a small per-die table, called GRAINS Scheduler Table (GST) in the internal DRAM, with one row per Strings page. Each row stores the bit address within the page (as marked by Offsets entries), the k -mer minimizer and its suffixes/prefixes (Fig. 9), a flag indicating whether the row is full (and if so, a pointer to an extension table), and a small one-hot encoded bitmap marking the target plane in the die. The bitmap allows GRAINS issue *multi-plane* SSD operations, so that multiple planes of the same die serve different accesses concurrently. When accessing Strings, GRAINS schedules requests by simply reading one row per GST sequentially and issuing requests across all dies and planes in a round-robin manner. This naturally coalesces accesses to the same page, avoids redundant page accesses, and achieves high parallelism with small metadata and no complex logic.

GRAINS can store GSTs in the internal DRAM because (i) its efficient address mapping (§8) frees most of the DRAM, and (ii) our compact k -mer representation (§6) keeps GST entries small. Internal DRAM is sufficient even for large read sets and graphs (e.g., our large evaluated read set with ~ 10 million query reads on a large-scale genome graph⁷ requires only 2.9 GB, well within the 4-GB internal DRAM in a typical 4-TB SSD). Furthermore, to ensure efficient functionality even when usage exceeds the internal DRAM's capacity, GRAINS keeps some $Sizes$ batches (§6) in host DRAM until the SSD finishes processing the current batch.

Fig. 11 shows an overview of Strings lookups. First, GRAINS's scheduler reads Strings addresses and corresponding k -mers from each GST (①) and sends the k -mers to the corresponding dies (②), incrementing each GST's row pointer for the next access. Second, GRAINS reads the corresponding Strings page, and after ECC_{LITE} (③), the IFP unit performs a comparison between the Strings in the indexed window and the incoming k -mer from the controller (④). Finally, GRAINS sends the comparison result and the unitig IDs of the matching k -mers back to the SSD controller.

⁷The graphs we evaluate (§11) are among the largest single graphs constructed in practice. Graph-based databases that encode even larger volumes of data typically consist of several graphs that need to be queried. This is because graph construction time scales super-linearly, making construction of a single very large graph prohibitively slow [342,425].

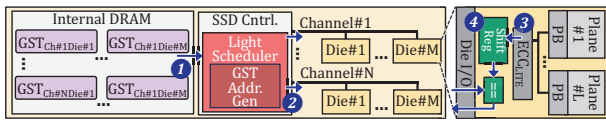


Fig. 11: Overview of GRAINS's Strings lookups.

7.3. Accessing Graph Colors

For each query k -mer that matches a unitig (§7.2), GRAINS retrieves the color of that unitig. Due to GRAINS's efficient scheduling (§7.2), unitig IDs are already retrieved in the internal DRAM in page order. Thus, given the underlying color encoding (§5), the unitigs' colors can be obtained via efficient sequential scans of the Color Bitmap (§5). Fig. 12 illustrates this. GRAINS's ISP units stream the next unitig ID from the internal DRAM into a small register (1) while concurrently scanning the Colors Bitmap from NAND flash dies (2). Given that the bitmap is consumed immediately, GRAINS's ISP units directly operate on NAND flash streams (as in prior work [420]), without buffering them in the internal DRAM. This avoids bottlenecking the internal DRAM's bandwidth. GRAINS uses only two small, 32-bit registers per channel: one for the current and one for the incoming bitmap chunk. As the bitmap is processed, each time a "1" is encountered (i.e., a new color is encountered), GRAINS increments *Color Index* (3). Once the bitmap position matches the unitig ID, GRAINS uses the Color Index to index the Colors (4). To access Colors, GRAINS uses the same approach as Offsets (§7.1), where, after light ECC (5), the IFP units select and transmit only the relevant portion of each page (6). The retrieved color entries are returned to the host, where each color identifies the database entries (e.g., species or samples) containing the matched sequence.

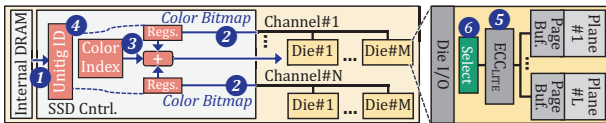


Fig. 12: Overview of GRAINS's Colors lookups.

7.4. Detailed Design of the IFP Processing Elements

GRAINS's IFP PEs follow a high-level flow: (i) a page is read into the die's page buffer, (ii) the page's content goes through ECC_{LITE}, and (iii) the PE operates on the page buffer contents.

IFP Hardware Units. GRAINS's IFP PEs implement two lightweight functions: (i) *selection*, which extracts a targeted window of a page, and (ii) *comparison*, which performs a bitwise match between a k -mer and a targeted window. The page buffer in a standard NAND flash die is already organized as a byte- or word-addressable latch-based array with existing sense amplifiers and column select logic. GRAINS's selection mechanism repurposes this column decoding circuitry. Given an offset, the column logic isolates the targeted bytes. Instead of streaming the full page over the channel bus, this targeted window is either routed directly to the die's I/O interface or fed internally into the IFP comparison unit. The comparison unit itself consists of a small shift register and a bitwise comparator that checks the k -mer against the targeted window.

Interaction with Flash Controllers. The flash controller sends a small parameter (an offset or a k -mer) to the on-die PE

via the existing channel bus. This parameter is stored in a small register on the die and is delivered alongside the standard page read command. After the IFP operations, the controller reads back only the result (instead of the full page).

8. FTL Integration and Reliability Support

GRAINS FTL requires only minor changes to the baseline FTL. It designates each block as either genomic or non-genomic, and identifies genomic accesses through GRAINS commands (§9). For all non-genomic data, vendor-specific FTL features remain untouched, and the SSD behaves conventionally. At the start of its SCC operations, since GRAINS does not perform writes during them, it flushes the standard L2P metadata to the flash and loads the much smaller GRAINS L2P metadata (as detailed below) into internal DRAM, while also keeping the other metadata of a conventional FTL.

Physical Allocation. Since GRAINS regularizes accesses, GRAINS's data placement is simple and enables (i) efficiently leveraging the SSD's internal bandwidth and (ii) minimizing mapping metadata. We uniformly distribute the blocks of the underlying graph data structures and the query reads across different channels, dies, and planes in a round-robin way. For efficient multi-plane operations, active blocks across different planes are aligned to the same page offset.

This placement works synergistically with GRAINS's batching and scheduling, enabling well-balanced, parallel accesses across all internal SSD resources. That said, for workloads with persistently skewed query distributions (e.g., when specific genomic regions are queried disproportionately often), profile-guided placement could offer additional benefits. Such an optimization would integrate seamlessly with GRAINS, affecting only the initial data layout. We describe this optimization further in an extended version [427].

FTL Metadata Size. GRAINS reduces the required FTL metadata size due to two reasons. First, instead of page-level L2P mappings (which, with 4 KiB pages, leads to ~ 4 GB of metadata for a 4-TB SSD), GRAINS stores L2P metadata at block granularity. This is possible because GRAINS performs no writes during SCC, and thus, no fine-grained remapping is needed. For a 1-TB graph with 12-MB blocks, this requires only 0.35 MB. Second, GRAINS's uniform data placement makes physical locations computable from base addresses and stride, minimizing per-block metadata needed for the SSD's reliable operations (e.g., read-disturbance counters). With per-block metadata, overall metadata size is 0.7 MB, freeing most of the internal DRAM for GRAINS's SCC and scheduling tables.

Error Correction. For data accessed by GRAINS's ISP units on the SSD controller (e.g., Color Bitmap), accesses occur after the regular ECC. For structures accessed by GRAINS's on-die hardware units, we adopt a lightweight on-die ECC technique, ECC_{LITE}, along with its associated *Bias Error Encoding* [322].

Other Management Tasks. Since in SCC mode, GRAINS does not incur writes, it does not require write-related management such as GC and wear-leveling. GRAINS performs other tasks for ensuring reliability before or after SCC since (i) the duration of each GRAINS process is significantly shorter than the manufacturer-specified threshold for reliable retention age

(e.g., a year [428]), and (ii) GRAINS avoids read-disturbance errors [290,291,362,363,377,380] during SCC since, due to its efficient execution, each block is read at most once. After SCC steps, to prevent read disturbance errors in the future, GRAINS refreshes blocks whose read count exceeds a specific threshold.

9. Interface Commands

GRAINS needs three new NVMe commands.

GRNS_Start initiates graph genome analysis acceleration. When it receives this command, GRAINS prepares itself for execution in SCC mode (§4). During this phase, GRAINS handles execution/data flow based on GRAINS FTL and FSM controller.

GRNS_Steps marks the start and end of each pipeline step and triggers GRAINS’s lightweight FSM on the SSD controller, which drives the internal execution during SCC. The host uses this command to signal that a query batch is ready, then transfers the batch to the SSD’s internal DRAM via the standard NVMe data path. Unlike conventional writes, these batches remain in the internal DRAM for processing and are not programmed to flash. The SSD then processes the batch through its internal pipeline with GRAINS’s FSM on the SSD controller coordinating execution and data flow, and returns results to the host. This enables pipelining multiple batches: as the SSD processes the current batch, the host prepares the next one. No additional commands are needed because the FSM fully determines the internal execution flow once a batch arrives. At the end of the last step, GRAINS returns to operation as a conventional SSD.

GRNS_Write is a specialized command for writing genome graphs to the SSD, which updates both the regular FTL’s L2P metadata and GRAINS’s small L2P metadata.

10. Deployment and Integration

GRAINS offers flexible deployment paths to minimize integration costs across both the hardware and software stacks due to three key reasons. First, as discussed in §4, GRAINS’s ISP/IFP steps can run on either specialized lightweight logic for maximum power efficiency, or on general-purpose ISP/IFP engines (e.g., [403,404,406–420]) for near-term adoption. Second, the required FTL modifications (§8) are minimal and non-disruptive, and the SSD behaves identically to a conventional SSD outside its SCC mode. Third, system-level integration is similarly seamless: GRAINS introduces only three new NVMe commands via standard vendor-specific extensions, and file system and driver changes are confined to recognizing genome graph files and dispatching these commands. We explain each of these reasons for GRAINS’s flexible deployment in more detail in an extended version [427].

11. Methodology

Performance. We design a simulator that models all of GRAINS’s components, including host operations, internal DRAM, accessing flash dies, in-storage hardware units (on flash dies and on the SSD controller), and SSD-host interfaces. We then feed the latency and throughput of each component to this simulator. Using this methodology, as also leveraged in prior SCC works (e.g., [301,313,315,369,392,429]), enables

us to flexibly incorporate state-of-the-art system configurations in our evaluations. For **hardware-based** components (e.g., when querying the graph): We implement GRAINS’s ISP/IFP logic units in Verilog and synthesize them with a 22nm library [430] using the Synopsys Design Compiler [431]. We use two state-of-the-art simulators, MQSim [367,432] to model SSD’s internal operations, and Ramulator [433,434] (new version introduced in [435,436]) to model internal DRAM. For **software-based** components (e.g., when preparing queries), we measure their performance on a real system with 1.5-TB DRAM (in all experiments unless mentioned otherwise) and AMD® EPYC® 7742 CPU [383] with 128 physical cores. We use SSD-G4 [385] and SSD-G5 [386] as described in §3 and Table 1 in our real system experiments. In our MQSim simulations for the ISP/IFP steps, we faithfully model these SSDs with configurations summarized in Table 1. For the software baselines, we measure performance using the best-performing thread counts on the same real system used for GRAINS’s software-based components. In an extended version [427], we provide further details regarding our methodology and how it manages correct timing composition between host-side and SSD-side time.

Area and Power. For GRAINS’s logic units, we use our Design Compiler synthesis results. SSD power is based on values of a Samsung 3D NAND flash-based SSD [382]. DRAM power is based on values from a DDR4 model [384,440]. For the CPU cores, we measure power with AMD µProf [441].

Evaluated Systems. We evaluate three key tasks: (i) k-mer set lookup, (ii) alignment-free read mapping, and (iii) alignment-based read mapping. We evaluate the following tools for analysis with large-scale genome graphs:⁸ (i) Fulgor [151] (FG), as a state-of-the-art node-centric software tool (which uses SSHash [341]), (ii) the MetaGraph framework [148,260–262] (MG) as a state-of-the-art edge-centric software tool; (iii) an *idealized* hardware-accelerated baseline (IdealAccMem) where all genome graph query operations execute *without* main memory overheads (i.e., zero latency, infinite bandwidth, infinite capacity). This configuration can represent an *ideal* PIM baseline as well. IdealAccMem still incurs the I/O overhead of bringing the large, low-reuse data from the storage system to main memory. Comparing against this baseline demonstrates the impact of storage I/O as a fundamental problem that persists even when main memory overhead is alleviated; (iv) GRAINS with all its optimizations, but with its lightweight hardware as an accelerator outside the SSD (GRN-Ext), to show the benefits of GRAINS’s

Table 1: Evaluated SSD configurations.

Specification	SSD-G4	SSD-G5
General	TLC NAND flash-based SSD 4 TB capacity, 4 GB internal DRAM [437]	
Bandwidth (BW)	PCIe Gen4 interface [438]; 7 GB/s sequential-read BW; 1.2-GB/s channel I/O rate	PCIe Gen5 interface [439]; 14.8 GB/s sequential-read BW; 2.4-GB/s channel I/O rate
NAND Config	16 channels, 8 dies/channel, 4 planes/die	

⁸As discussed in §2.2, DBGs scale substantially better than other genome graph structures as the genetic diversity in the database increases. Other tools (e.g., VG [442] and minigraph [354]) that use variation graphs as their graph structure do not scale as well and, hence, are not typically used in scenarios with large-scale, genetically diverse databases [148,338].

optimizations without ISP/IFP, but with more storage-friendly execution/data flow. The lightweight hardware is connected to the host system via a PCIe interface with 16 GB/s bandwidth, and leverages the host DRAM to store its scheduling meta-data (§7.2). Similar to GRAINS’s full implementation, GRN-Ext performs input query processing on the host (§6), but instead of relying on its ISP/IFP logic units to query the graph (§7), it uses the same lightweight accelerator logic units outside the SSD; (v) GRAINS’s full implementation, including ISP and IFP (GRN). All these tools achieve the same accuracy since their underlying graphs are *lossless* encodings of the database’s k-mer sets [148,151,260–262]. For the alignment-based mapping, we integrate all tools with SeGraM, a state-of-the-art sequence-to-graph alignment accelerator [231].

Datasets. We build graphs as described in §3. The graph (with colors) is 659 GB for FG, GRN, and IdealAccMem, and 822 GB for MG. We evaluate query read sets with 10M (QL), 1M (QM), and 100K (QS) reads. Note that smaller queries are also critical, as detailed in §3.

12. Evaluation

12.1. Performance

K-mer Set Lookup. Fig. 13 shows the speedups of different systems over FG for k-mer set lookups. We make three key observations. First, GRAINS provides significant speedups. In systems with SSD-G4 (SSD-G5), GRN provides $8.4\times$ ($5.5\times$) and $11.9\times$ ($8.5\times$) average speedups over FG and MG, respectively. Second, by making the execution/data flow more storage-friendly, GRAINS’s implementation outside the SSD also provides large benefits. GRN-Ext provides to $3.4\times$ and $5.0\times$ average speedup over FG and MG, respectively, across all inputs and SSDs. Third, GRAINS’s full implementation provides significantly larger benefits over GRAINS’s implementation outside SSD. GRN provides $2\times$ average speedup over GRN-Ext. While GRN-Ext exploits locality across queries by merging requests to the same regions of each graph structure and improving access patterns, the locality it captures cannot amortize the cost of transferring the large volume of low-reuse data across the storage-to-host interface. Thus, GRAINS’s full implementation provides additional, significant benefits by alleviating the large I/O overheads through its SCC design. We further evaluate the execution time breakdown of different GRAINS steps in an extended version [427].

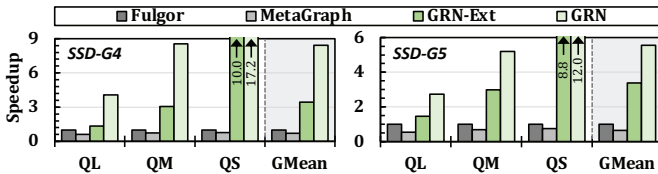


Fig. 13: Speedups with different input query sets and SSDs.

Impact on Cost Efficiency. GRAINS improves system cost-efficiency since it analyzes large amounts of data in the SSD and does *not* rely on either large host DRAM or high host-SSD interface bandwidth. We show this by evaluating two systems: a cost-optimized system (\$) with 64-GB host DRAM and SSD-G4, and a costlier, performance-optimized system (\$\$\$) with 1.5-TB host DRAM and SSD-G5. Fig. 14 compares

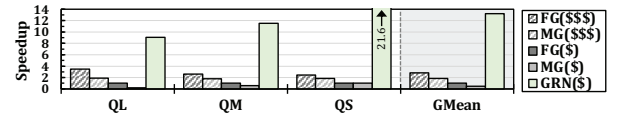


Fig. 14: Speedup of GRAINS on a low-cost system, i.e., GRN(\$), over baselines on low-cost and performance-optimized systems.

GRAINS on the cost-optimized system with the baselines on both systems (with a representative query set, QM). To fairly evaluate the baselines when host DRAM is smaller than the graph, we reduce I/O overheads as much as possible in software for this scenario. To this end, we partition the graph such that each subgraph fits in the host DRAM, so random accesses to the graph do not repeatedly access the SSD. We partition the graph using database entry metadata to guide partitioning choices, placing entries that likely share many k-mers into the same partition. Despite the benefits of this optimization, two sources of overhead remain: (i) the I/O overhead of transferring subgraphs, and (ii) the need to query against each subgraph. We make two observations. First, GRAINS on the cost-optimized system significantly outperforms the baselines even on the performance-optimized system. GRN(\$)\$ provides $4.7\times$ and $5.2\times$ average speedup over FG(\$\$\$) and MG(\$\$\$), respectively. Second, the performance of baselines on the cost-optimized system suffers significantly. When compared on the same cost-optimized system, GRN(\$)\$ provides $13.2\times$ and $26.9\times$ average speedup over FG and MG, respectively. We conclude that GRAINS improves both system cost-efficiency and system performance, which is critical for facilitating (i) the wide adoption of graph-based genome analysis, and (ii) analysis on low-cost, portable devices, a scenario rising in importance with the development of portable sequencing devices [443–446] for on-site genome analysis [159,160,447,448].

Effect of Different GRAINS Optimizations. Fig. 15 demonstrates the benefits of different GRAINS optimizations. GRN-B is a GRAINS configuration including only the batching optimization (§6). GRN-B-S is a GRAINS configuration including both the batching and scheduling (§7.2) optimizations (equivalent to GRN-Ext, §11). GRN-B-S-SCC is GRAINS’s full configuration, including its SCC. Please note that we do not show a configuration with naive SCC and without batching and scheduling. This is because simply using SCC without these techniques leads to performance loss due to the irregular access patterns to genome graphs, which are inefficient due to limitations of the SSD’s hardware (e.g., costly conflicts in internal SSD resources such as channels and NAND flash chips [319–321] during random and dependent accesses). We make three observations. First, GRN-B provides $1.5\times$ and $2.2\times$ average speedup over FG and MG, respectively. This is due to more efficient accesses to the Offsets data structure. Second, GRN-B-S provides $2.3\times$ average speedup over GRN-B as it enables more

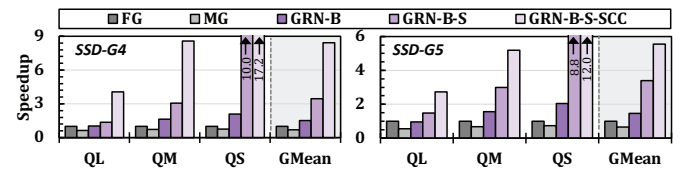


Fig. 15: Ablation tests of different GRAINS optimizations.

storage-friendly accesses to Strings and Colors as well. Finally, through its efficient SCC, GRN-B-S-SCC provides $2.0\times$ and $4.6\times$ average speedup over GRN-B-S and GRN-B, respectively.

Effect of the Number of SSDs. GRAINS can benefit from multiple SSDs in two ways: First, different graphs can be stored on separate SSDs and analyzed concurrently, each benefiting from GRAINS (as shown in Fig. 13). Second, a single graph can be partitioned across SSDs to increase throughput. This is enabled by GRAINS’s execution flow supporting efficient graph partitioning, as detailed in an extended version [427]. We evaluate this scenario in Fig. 16. We observe that GRAINS maintains large speedups. This is because, as the external bandwidth increases for the baselines with more SSDs, the internal bandwidth also increases. Although there is a slight decrease in speedup when moving to two SSDs in QS, the speedup is still significant ($14.2\times$ and $9.4\times$ with SSD-G4 and SSD-G5). This decrease is because, after increasing internal bandwidth, the impact of software query processing becomes relatively larger. Therefore, we can further accelerate these steps (e.g., with a sorting accelerator) to increase benefits even further. We conclude that GRAINS can efficiently leverage multiple SSDs, making it also suitable for distributed systems.

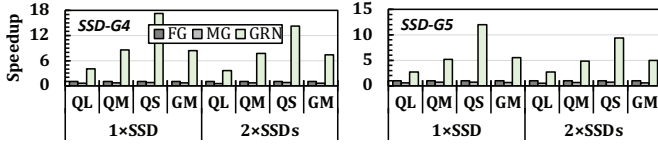


Fig. 16: Speedups with different numbers of SSDs.

Comparison to IdealAccMem. Fig. 17 shows speedups over IdealAccMem. Despite IdealAccMem’s benefits, it still incurs I/O accesses to transfer data to main memory, even when DRAM is larger than data size (as in our evaluation). Through its storage-aware design, with efficient ISP/IFP operations, we observe that on the system with SSD-G4 (SSD-G5), GRAINS provides $7.4\times$ ($4.3\times$) average speedup over IdealAccMem.

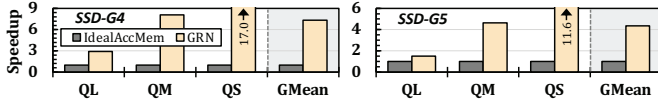


Fig. 17: Speedup over IdealAccMem.

Alignment-Free Read Mapping. Fig. 18 shows speedups over FG when performing alignment-free read mapping. In this case, as discussed in §6, GRAINS sends the colors of all k-mers to the host system, and the host system determines the colors of each read based on the colors of its k-mers. We observe that on systems with SSD-G4 (SSD-G5), GRN provides $7.9\times$ ($5.2\times$) and $11.2\times$ ($8.0\times$) average speedups over FG and MG, respectively.

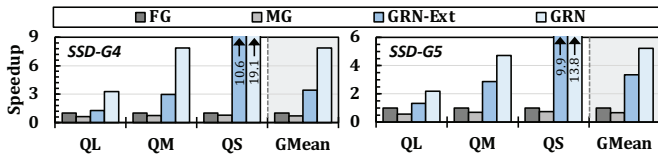


Fig. 18: Speedups for read mapping.

Alignment-Based Mapping. We integrate GRN, FG, and MG with SeGraM [231] to perform alignment on the candidate

graph regions identified by each tool (we use the throughput of alignment operations reported by SeGraM [231]). Fig. 19 shows the speedups of different systems over FG +SeGraM. We observe that SeGraM+GRN provides $6.2\times$ and $9.0\times$ average speedup over SeGraM+FG and SeGraM+MG.

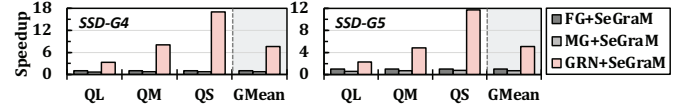


Fig. 19: Speedups for read mapping with alignment.

12.2. Area and Power Overheads

Table 2 summarizes area and power for GRAINS’s logic units at 333MHz. While these units can operate at higher frequencies, their throughput is already sufficient since GRAINS’s ISP/IFP pipelines are bottlenecked by flash reads. We use area/power values of ECC_{LITE} from the original work [322].⁹ GRAINS’s hardware overhead is very small. On-controller units are 0.0025 mm^2 (0.7% of the four ARM cores [450] on an SSD controller). GRAINS’s on-die logic area is mainly (99%) for ECC_{LITE}, which, as reported by [322], takes only 0.2% of the flash chip area and fits well within its power budget.

As detailed in §7, GRAINS’s SCC can execute on either (i) our lightweight, specialized ISP/IFP units or (ii) general-purpose ISP (e.g., [406–420]) and IFP (e.g., [403,404]). Choosing between these GRAINS configurations is a design tradeoff: general-purpose units facilitate earlier adoption, while specialized units offer better area and power efficiency. For example, GRAINS’s ISP units on the SSD controller consume two orders of magnitude lower area and $60.1\times$ lower power than hardware units of a state-of-the-art general-purpose ISP.¹⁰

Table 2: Area and power consumption of GRAINS’s logic units.

Logic unit	Area [mm^2]	Power [mW]
On SSD Controller	0.0025	0.21
On Die (ECC _{LITE})	0.036	18.04
On Die (Others)	0.000093	0.01

12.3. Energy

We find the energy consumption of each evaluated graph-based genome analysis tool (when performing k-mer set lookups) based on the energy consumption of different components (e.g., host processor, host DRAM, host-SSD communication, SSD components, and logic units). We calculate each component’s energy based on its dynamic/idle power and execution time. We observe that across our evaluated input queries and SSDs, GRAINS provides $4.4\text{--}21.0\times$, $12.2\text{--}31.6\times$, and $3.1\text{--}20.7\times$ energy reduction over FG, MG, and IdealAccMem, respectively, when performing k-mer set lookups.

13. Related Work

To our knowledge, GRAINS is the first storage-centric computing (SCC) system designed to significantly reduce I/O

⁹Since we do not have access to the larger technology node used in [322], we scale ECC_{LITE}’s area to the lower node with the methodology in [449].

¹⁰Note that the power/area of GRAINS’s IFP units are mainly dominated by ECC_{LITE}, needed in both GRAINS and other IFP systems (e.g., [322,403]).

data movement overheads and improve end-to-end performance, energy-efficiency, and cost-effectiveness of graph-based genome analysis.

Analysis with Genome Graphs. Many works (e.g., [265,331,344–354]) propose software tools for genome graphs. Several works propose hardware tools for querying genome graphs by alleviating computation (e.g., [231–240]) and/or memory overheads (e.g., [231,241–244]). However, these works do not alleviate storage I/O overheads, whose impact on end-to-end performance becomes even larger as other overheads are alleviated. As shown in Fig. 19, GRAINS can flexibly integrate with these designs to alleviate their I/O overhead. Some works (e.g., [245–257]) accelerate genome graph *construction*, which is an important, but orthogonal task.

Accelerating Graph Analysis. Many works accelerate graph analysis by alleviating computation (e.g., [360,451–466]), main memory (e.g., [465,467–481]), or I/O (e.g., [297–305]) overheads. However, they are neither sufficient for nor target the unique demands of genome graphs (as detailed in §2.3).

Accelerating Genome Analysis. Many works (e.g., [116,158–203,205–214,217,225,227,229,313,315,482–511]) propose hardware optimizations for conventional genome analysis on linear sequences, including SCC systems (e.g., [204,241,313–318]) for these analyses. However, none of them are suitable for graph-based genome analyses due to fundamentally different data structures, execution flow, and irregular and dependent access patterns.

Particularly, GRAINS exploits the properties of graph-based genome analyses in three ways. First, since GRAINS needs to handle *dependent accesses across large data structures residing in the SSD*, it needs to handle scheduling directly in the SSD. To do so, GRAINS introduces a novel data/execution flow and lightweight scheduling (§7), fundamentally different from prior approaches that regularize accesses purely through query scheduling in the host [315] or via offline preprocessing [313]. Second, GRAINS’s batching and task partitioning efficiently exploit the underlying properties of genome graphs (§6), which no prior work considers. Third, GRAINS performs end-to-end sequence-to-graph read mapping (which is more complex than traditional sequence-to-sequence mapping).

Several works propose techniques for batching k-mer queries to improve access patterns to genomic data structures (e.g., [512–514]). Despite their benefits, these techniques cannot be directly adopted in SCC designs for genome graphs due to the dependent and random accesses to different data structures. A purely host-side approach would require a separate round trip to the SSD for each dependent access stage, undermining the locality benefits of batching. On the other hand, directly adopting these techniques within the SSD stresses the limited available hardware resources in modern SSDs. GRAINS addresses these limitations with a co-designed approach. First, we perform lightweight, genome-graph-aware batching on the host (exploiting the small Sizes metadata in the host, §6). Second, once these batched requests are dispatched to the SSD, our new, lightweight in-storage scheduling orchestrates all subsequent dependent accesses to large data structures inside the SSD, without any further round-trips to the host (§7).

SCC Designs. Various works propose SCC for different applications (e.g., in AI/ML [299,300,309,322,392,393,396,397,515–523], string matching and read mapping [204,313,316,317,524], k-mer counting [314], graph analytics [296,297,306], and others [307,405,525–545]). Several works propose general-purpose SCC in storage [406–421,429,546], SSDs integrated closely with FPGAs [395,547–552], or with GPUs [553]. None of these works targets analysis with genome graphs.

Some works (e.g., [554,555]) propose performing sub-page reads to avoid unnecessary data movement from the flash dies. However, they focus solely on reducing data transfer granularity and do not perform computation within the dies. Several works propose performing computation in the dies [322,403,404], or using flash memory [369,405,556,557]. However, directly applying these techniques to genome graph analysis is undesirable due to the irregular, data-dependent accesses, which hinders leveraging die-level parallelism. GRAINS addresses this challenge through its genome-graph-aware algorithm-architecture co-design.

Several works (e.g., [397,421,558]) propose SCC designs that combine ISP, IFP, and PIM. However, these works are not sufficient to handle the unique properties of graph-based genome analysis, which pose challenges absent from prior SCC work. We address these challenges via our genome-graph-aware query reordering techniques and lightweight scheduling, enabled via GRAINS’s holistic co-design of data/execution flow, data layout, and FTL integration.

14. Conclusion

We introduced GRAINS, the first storage-centric system designed to alleviate the I/O overhead of analysis with large genome graphs. Through detailed examination of these analysis pipelines, we (i) make them more storage-friendly and (ii) enable efficient in-storage and in-flash processing. Our evaluations show that GRAINS improves performance, energy efficiency, and system cost-effectiveness of graph-based genome analysis at low area and power costs. We hope that GRAINS’s optimizations will inspire and benefit storage-centric computing systems in other domains to likewise improve their performance and energy efficiency without relying on expensive hardware resources.

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